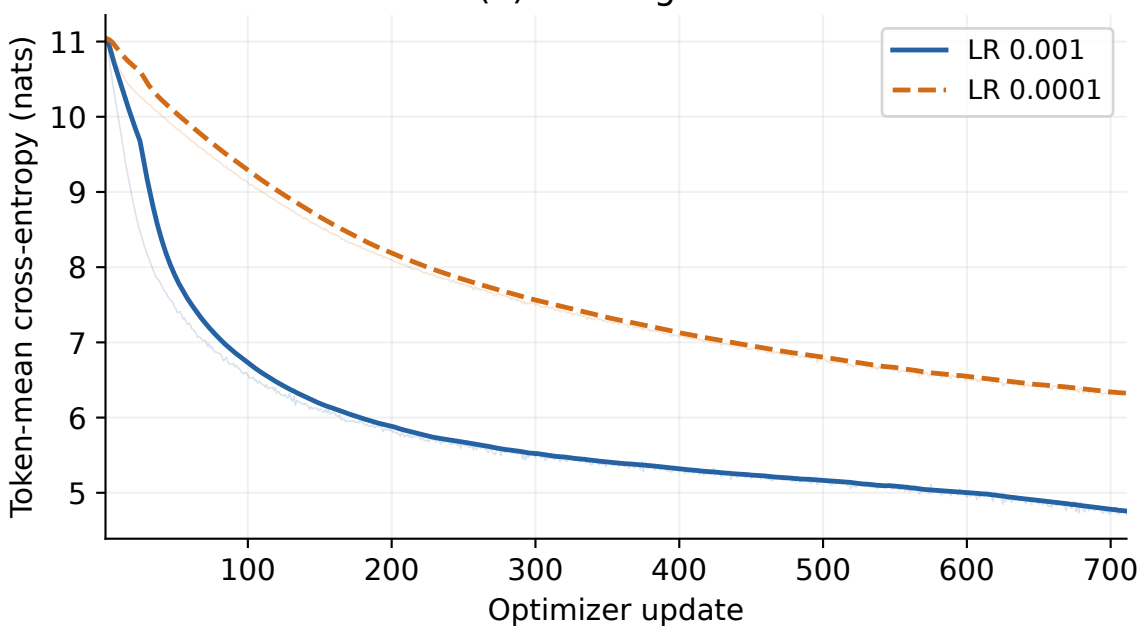
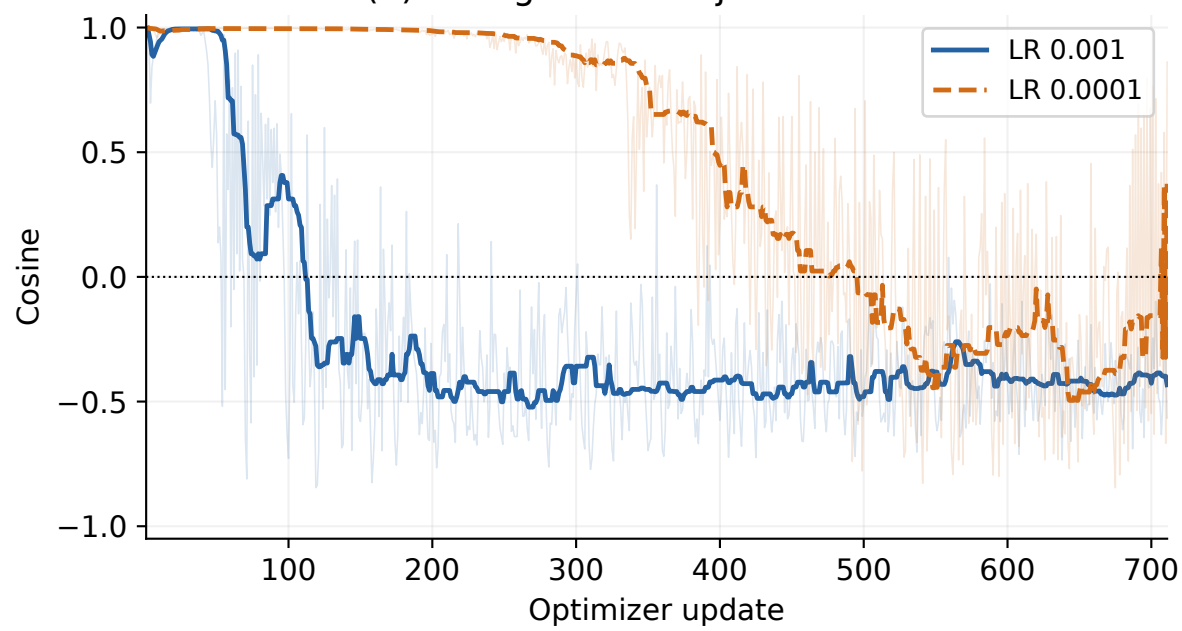
Pythia-14M: comparison across learning rates
Warmup only; matched initial weights and data order; seed 1234

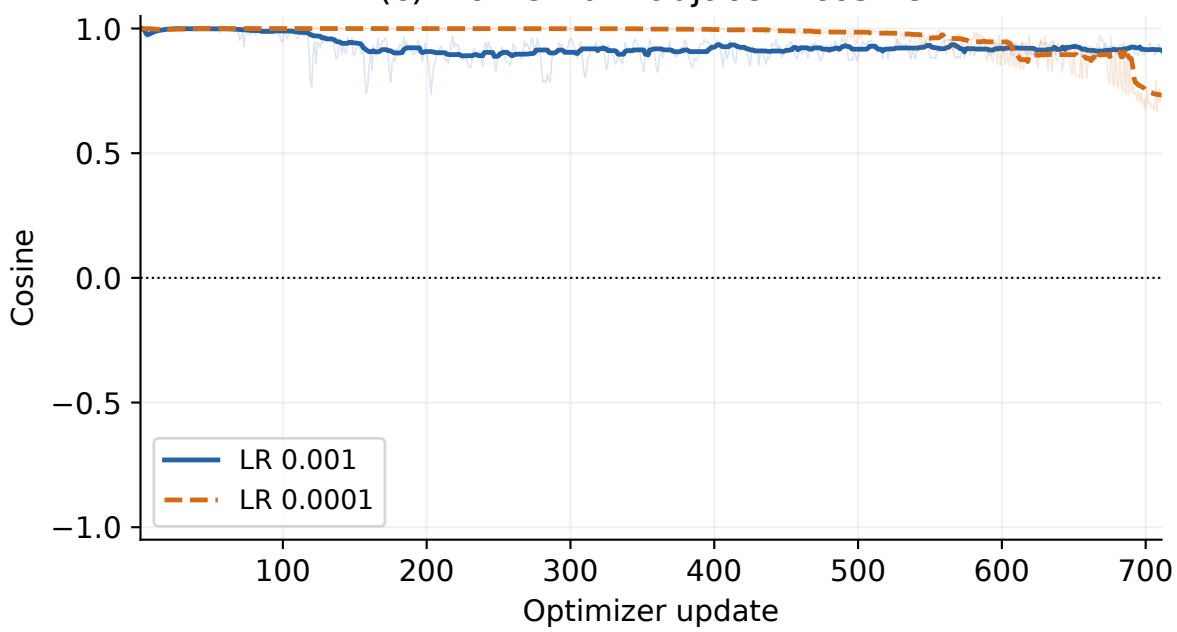
(a) Training loss



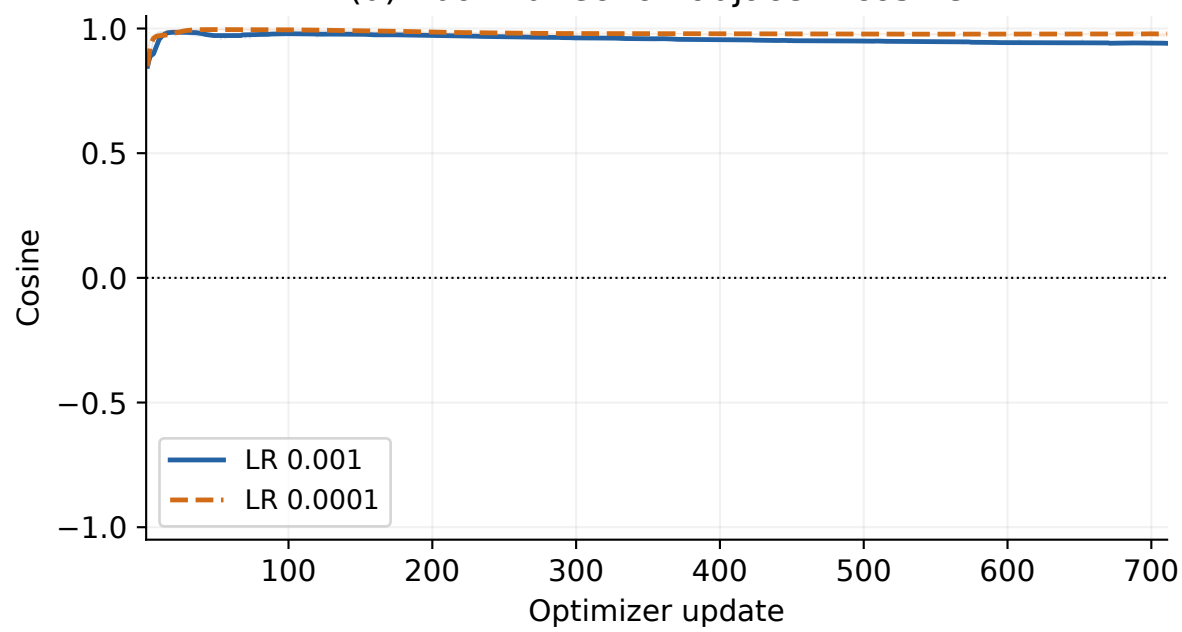
(b) Raw-gradient adjacent cosine



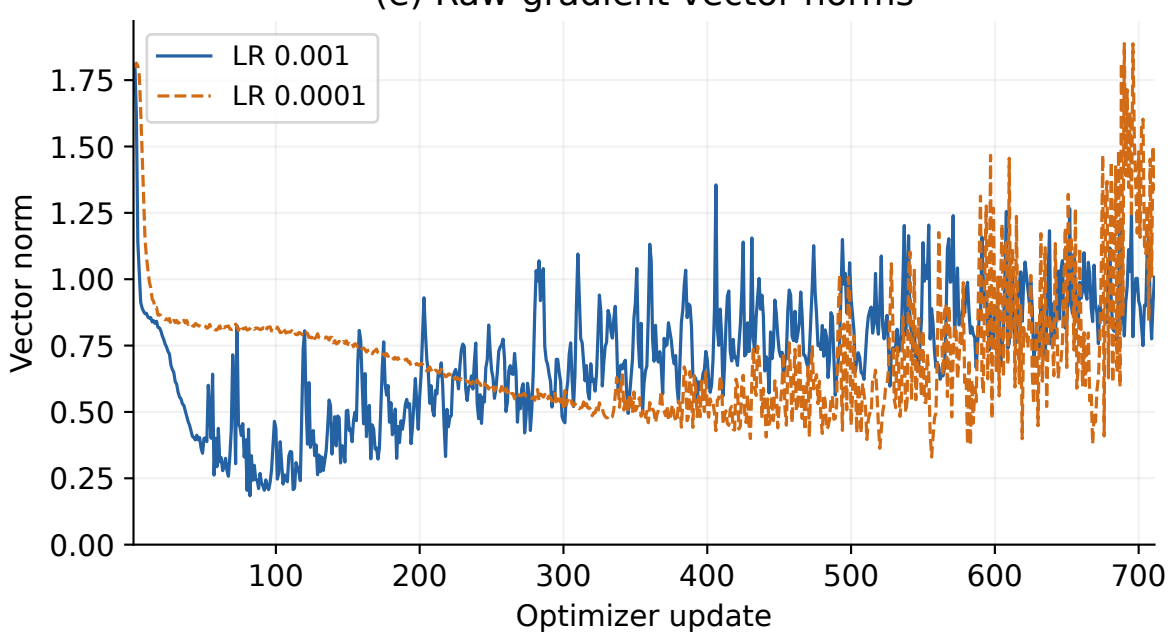
(c) Momentum adjacent cosine



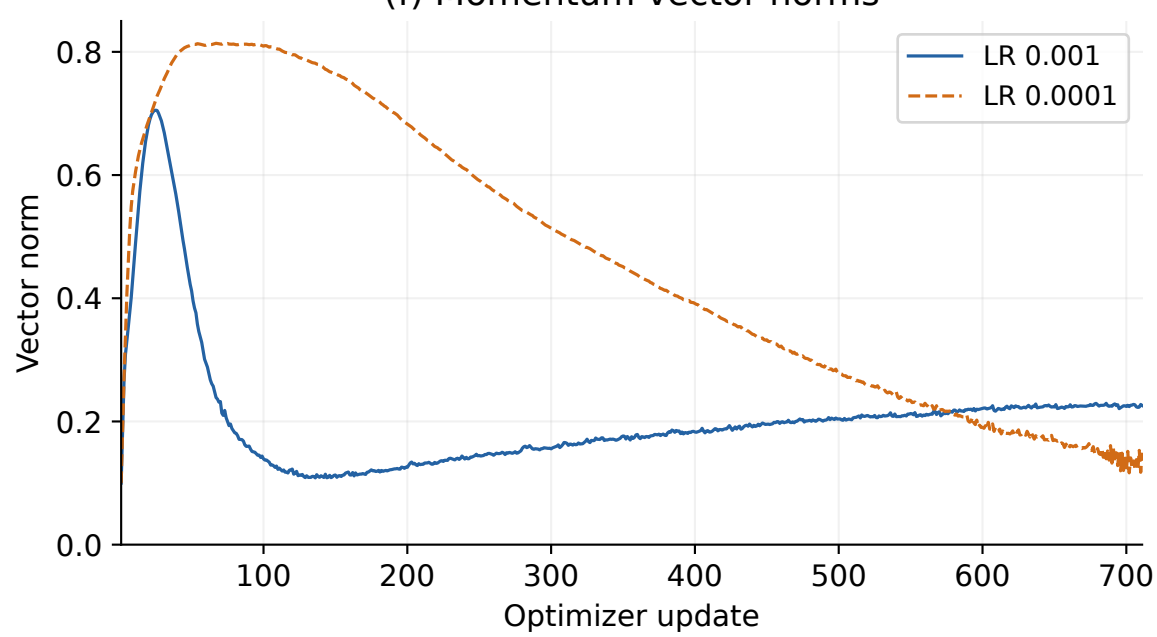
(d) Adam-direction adjacent cosine



(e) Raw-gradient vector norms



(f) Momentum vector norms



Light loss/cosine traces: individual updates. Bold loss: trailing 25-update token-weighted mean; bold cosines: trailing 25-update median.
Norms are unsmoothed and use linear axes. All cosine panels use the same range. Display: full-sized updates 1-711.
m is stored uncorrected Adam exp_avg; d is the bias-corrected adaptive direction. Partial update 712 remains in numerical results.